



Getting started: 60 seconds to Underworld

Rohan Byrne

DOI <https://doi.org/10.6084/m9.figshare.33193419>

Abstract

As the old truism states, the best way to learn is by doing. This guide can get you from zero to hero in 60 seconds - give or take.

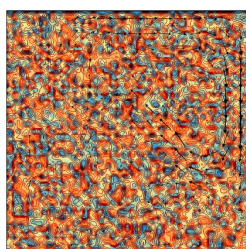
Keywords Documentation

As the old truism states, the best way to learn is by doing.

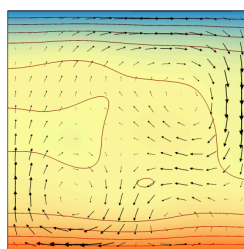
This is terribly unhelpful since, really, you can't 'do' anything *until* you learn. This is a [Catch-22](#) that every grad student encounters; doubly so when it comes to anything involving code, where daily life provides very little in the way of intuition.

Unlike other geodynamic modelling software, **Underworld** is designed to be approached with code. The downside is that the learning curve starts with a bump. The enormous upside is that, once you have a little fluency under your belt, you will find your science moves at the speed of thought.

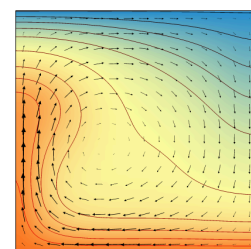
But how do you surmount that bump?



(a)



(b)



(c)

Figure 1: If all goes well, this is where you'll be five minutes from now!

1. PREPARING YOUR WORKSPACE

In this short how-to, we're going to lay out a 60-second scheme to get you up and running in **Underworld**. But before we start, we'll need to make sure you've got the right tools.

You will need:

- A computer. Although Underworld is proudly [Turing Complete](#), hand iteration is not recommended.
- A terminal. On Windows this might be [PowerShell](#) or [Command Prompt](#); on Mac and Linux it's just called the [Terminal](#). In any case what you'll get is a blank background with a blinking cursor beckoning you to type your bidding into the computer.
- A working instance of [Docker](#). Docker supports [containers](#) which are like mini-computers that live inside a bigger computer. If you're new to containers, try Docker Desktop: [get started here](#).
- A browser. We recommend [Chrome](#) for ease of use or [Firefox](#) if you're concerned about the depredations of late capitalism. Avoid [Safari](#) (both the software and the [kind that involves lions](#)).
- Some place on your hard drive that you're happy to use for Underworld stuff: a folder in your Documents, for instance. Anywhere will do so long as it's not on a network or share drive (this could make your [sysadmin](#) unhappy). Jot down the filepath of this directory as you will need it shortly. Try to get the 'absolute' filepath if you can (on Windows, this will start with `C:\` or similar; on Linux and Mac, simply `/`).

That's it!

2. TYING YOUR SHOELACES

So we don't get tripped up by any lurking [gremlins](#), let's quickly test to see if things are working as they should.

1. Open your terminal. (Try your OS's search bar if you can't find it)
2. Type in `docker` and hit enter. You should get a little introduction and a list of commands: this is everything that Docker can do for you.
3. Type in `docker run hello-world` and hit enter. After a little number-crunching you should receive a cute message from the good folks at Docker Hub.

```
rsbyrne@Mars: /c/Users/rohan/Documents/PhD_Mars
rsbyrne@Mars:/c/Users/rohan/Documents/PhD_Mars$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

rsbyrne@Mars:/c/Users/rohan/Documents/PhD_Mars$ _
```

Figure 2: Thanks Docker !

If you get errors instead, you will have to do some good old [troubleshooting](#). None of the rest of our guide will work unless Docker is installed and correctly configured.

But if everything looks [shipshape](#), you're ready to run.

3. ON YOUR MARKS...

Take a deep breath. You are about to become a champion Underworld2 geodynamic numerical modeller.

4. Go!

Start the clock.

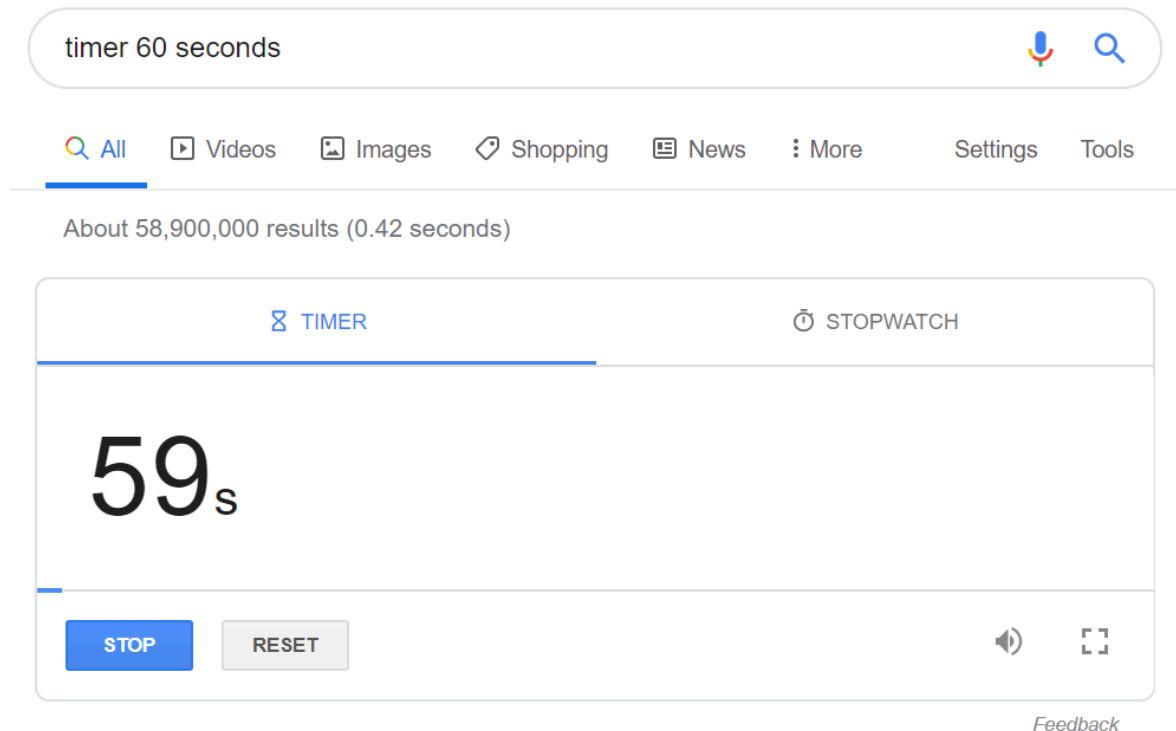


Figure 3: Hardly seems fair...

1. Open your terminal
2. Do you have your target directory from before? You'll need it. Type in the following docker command, where 'YOUR_DIRECTORY' is your nominated folder:

```
docker run -v YOUR_DIRECTORY:/home/jovyan/workspace/ -p 127.0.0.1:8888:8888 underworldcode/underworld2:latest
```
3. Stop the clock! This may take some time depending on your internet connection and whether you've tried it before. When it's ready, you should get a message with a URL in it.
4. Go to your browser and type the following into the address bar: 127.0.0.1:8888 You may need to provide a 'token' to access: you can copy-paste this from the command prompt where you ran Docker just before. Consider following the prompts to set up a password, which will save you time in the future.
5. You are now in 'Jupyter Notebooks', which is a browser-based environment for writing and running code. Navigate to the 'workspace' directory. This directory is actually the very same as YOUR_DIRECTORY - it has been 'mounted' into the container. Make a new file in Jupyter and you should see it pop up in the appropriate place on your desktop, and vice versa.
6. Inside 'workspace', click the 'new' dropdown button and select 'Python 3'. It will open a new 'notebook' with a blank 'cell' at the top.

```
rsbryne@Mars: /c/Users/rohan/Documents/PhD_Mars$ docker run hello-world
Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

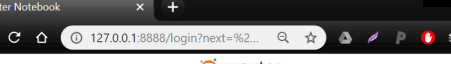
For more examples and ideas, visit:
 https://docs.docker.com/get-started/

rsbryne@Mars: /c/Users/rohan/Documents/PhD_Mars$
```

(a)

[illegible]

(b)



Jupyter Notebook

127.0.0.1:8888/login?next=%2F

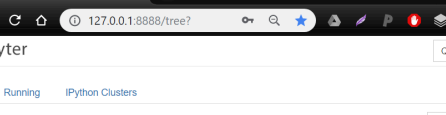
Jupyter

Token authentication is enabled

If no password has been configured, you need to open the notebook server with its login token in the URL, or paste it above. This requirement will be lifted if you [enable a password](#).

Password or token: Log In

(c)



The screenshot shows the JupyterLab interface. On the left, the 'Files' panel displays a file browser with a list of folders and files. The 'workspace' folder is highlighted with a red rectangle. The main area shows a terminal window with the command 'python3 test.py' and its output.

(d)

The screenshot shows a Jupyter Notebook interface. The top bar includes a browser address bar with the URL '127.0.0.1:8888/notebooks/wo...' and a Jupyter logo. Below the browser bar is the Jupyter Notebook header with the title 'Untitled1' and a menu bar with options: File, Edit, View, Insert, Cell, Kernel, Widgets, Help. The main area of the notebook displays the output of two commands: `import underworld as uw` and `help(uw)`. The output shows the help text for the 'underworld' package, which describes it as a python-friendly version of the Underworld geodynamics code, providing a programmable and flexible front end to all the functionality of the code running in a parallel HPC environment. It lists advantages such as access to the power of python libraries for setup of complex problems, analysis at runtime, problem steering, and coupling of multiple problems. It also mentions that 'underworld2' is integrated with the 'literate programming' environment of 'underworld'.

```
In [1]: import underworld as uw

In [2]: help(uw)

Help on package underworld:

NAME
    underworld

DESCRIPTION
    Underworld2 is a python-friendly version of the Underworld geodynamics
    code which provides a programmable and flexible front end to all the
    functionality of the code running in a parallel HPC environment. This
    gives significant advantages to the user, with access to the power of
    python libraries for setup of complex problems, analysis at runtime,
    problem steering, and coupling of multiple problems.

    Underworld2 is integrated with the literate programming environment of
```

(e)

Figure 4: Being able to imagine a job is 9/10ths of being able to perform it (but the remaining 1/10 is what you get paid for...)

Believe it or not, you now have a fully functional geodynamic numerical modelling environment at your very fingertips!

Still time on the clock? Let's see if we can get an actual model up and running. At the bottom of this article you will find a long code snippet. This is a complete, working, research-ready Underworld model which demonstrates the consequences of exponentially temperature-dependent rheology. Just **cut, paste, and hit enter**.

You could be using this for science right now: what are you waiting for?

5. PODIUM FINISH

If everything goes right, this guide should have taken you from [Spider Solitaire](#) to geodynamic modeller *par excellence* in 60 seconds. But if it didn't, you can always [get in touch with us](#) for help. **Underworld** is more than just a bit of software: it's a vibrant, passionate, and diverse research community dedicated to helping you answer your question - and have fun doing it.

Now you've taken the first step, try getting started on our interactive [user guide](#), which can be found online as well as in the container itself, in the root directory. And follow our channels on [Ghost](#), [Medium](#), [Facebook](#), [Twitter](#), and [GitHub](#) for more how-tos, tutorials, meet-ups, updates, and general research chatter.

Turns out the best way to learn really is by *doing*. But that doesn't mean you have to do it alone.

-Rohan

```
# MY FIRST MODEL:
# ARRHENIUS RHEOLOGY IN UNDERWORLD
...

Cut and paste me into one or more cells
of you Jupyter notebook and hit enter.
Or save as a script and execute from terminal
with:
mpirun -n 4 python YOUR_SCRIPT.py
...to execute in parallel on four cores.
Change the parameters to investigate
the impact of varying degrees of
temperature-dependent viscosity.
...

import numpy as np

import underworld as uw
fn = uw.function
import glucifer

# Define physical parameters:
Ra = 1e7
eta0 = 3e4
res = 64

# Create the computational mesh:
mesh = uw.mesh.FeMesh_Cartesian(
    elementRes = (64, 64)
)

# Create the system variables:
temperatureField = mesh.add_variable(1)
temperatureDotField = mesh.add_variable(1)
pressureField = uw.mesh.MeshVariable(mesh.subMesh, 1)
velocityField = mesh.add_variable(2)

# Define the walls:
```

```

inner = mesh.specialSets["MinJ_VortexSet"]
outer = mesh.specialSets["MaxJ_VortexSet"]
left = mesh.specialSets["MinI_VortexSet"]
right = mesh.specialSets["MaxI_VortexSet"]

# Define boundary conditions along the walls:
tempBC = uw.conditions.DirichletCondition(
    variable = temperatureField,
    indexSetsPerDof = (inner + outer)
)
velBC = uw.conditions.DirichletCondition(
    variable = velocityField,
    indexSetsPerDof = (left + right, inner + outer)
)

# Define the laws of physics:
buoyancyFn = Ra * temperatureField * (0., 1.)
viscosityFn = fn.math.pow(
    eta0,
    1. - temperatureField
)
heatingFn = 1.
diffusivityFn = 1.

# Create the mathematical systems that solve them:
stokes = uw.systems.Stokes(
    velocityField = velocityField,
    pressureField = pressureField,
    conditions = [velBC,],
    fn_viscosity = viscosityFn,
    fn_bodyforce = buoyancyFn
)

solver = uw.systems.Solver(stokes)

advDiff = uw.systems.AdvectionDiffusion(
    phiField = temperatureField,
    phiDotField = temperatureDotField,
    velocityField = velocityField,
    fn_diffusivity = diffusivityFn,
    conditions = [tempBC,]
)

# Define the basic running constants:
step = fn.misc.constant(0)
modeltime = fn.misc.constant(0.)

# Make a function to initialise the model:
def initialise():
    step.value = 0
    modeltime.value = 0.
    randArray = np.random.rand(*temperatureField.data.shape)
    temperatureField.data[...] = randArray
    temperatureField.data[outer] = 0.
    temperatureField.data[inner] = 1.
    solver.solve()

```

```

# Create a figure so we can see what's happening:
fig = glucifer.Figure()
fig.append(
    glucifer.objects.Surface(
        mesh,
        temperatureField,
        colourBar = False
    )
)
fig.append(
    glucifer.objects.VectorArrows(
        mesh,
        velocityField
    )
)
fig.append(
    glucifer.objects.Contours(
        mesh,
        fn.math.log2(viscosityFn),
        colours = 'red black',
        colourBar = False
    )
)

# Make a convenience function for the figure:
def show():
    if uw.mpi.rank == 0:
        print('step: ', step.value)
        print('time: ', modeltime.value)
    fig.show()

# Let's have a look at what we've got:
initialise()
show()

# Make a function that iterates the model:
def iterate():
    dt = advDiff.get_max_dt()
    advDiff.integrate(dt)
    solver.solve()
    step.value += 1
    modeltime.value += dt

# Make a convenience function for multiple iterations:
def go(n):
    for i in range(n):
        iterate()

# Let's go!
for i in range(100):
    go(10)
    show()

```